

Irreducible Representations of $\mathfrak{sl}(2, \mathbb{C})$ and Vector Fields on $\mathbb{R}^2$ Induced by Möbius Transformations

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Abstract

This paper presents a comprehensive study of the irreducible finite-dimensional representations of the complex Lie algebra $\mathfrak{sl}(2, \mathbb{C})$. We begin with a foundational overview of Lie algebras and their modules, highlighting key concepts such as nilpotent maps, weight spaces, and Schur's Lemma. Building on this framework, we construct the family of representations V_d , arising from the action of $\mathfrak{sl}(2, \mathbb{C})$ on homogeneous polynomials in two variables of degree d , and prove their irreducibility. We then demonstrate that every finite-dimensional irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ is isomorphic to one of these modules. In the latter part of the paper, we explore the geometric implications of these representations by examining the vector fields on $\mathbb{R}^2$ generated via exponentiation and Möbius transformations.

1 First Definitions

Definition: Let F be a field. A *Lie Algebra* over F is an F -vector space L , equipped with a bilinear map called the Lie bracket

$$\begin{aligned} L \times L &\rightarrow L \\ (x, y) &\mapsto [x, y] \end{aligned}$$

that satisfies the following properties

- $[x, x] = 0, \forall x \in L$
- $\sum_{\text{cyclic}} [x, [y, z]] = 0$ (Jacobi Identity)

The Lie bracket is bilinear and so $[x, y] = -[y, x]$.

Definition: Let V be a vector space. Turn V into a Lie algebra by defining $[x, y] = 0$ for all $x, y \in V$. Such a Lie algebra is called an *abelian Lie algebra*.

Example: Let V be a finite dimensional vector space. Set $gl(V) := \{T : V \rightarrow V : T \text{ linear}\}$. We turn $gl(V)$ into a Lie algebra by defining the Lie bracket as

$$[x, y] = x \circ y - y \circ x$$

for all $x, y \in gl(V)$.

Definition: Let L be a Lie algebra. A *Lie subalgebra* of L is a vector subspace $K \subseteq L$ such that $[x, y] \in K$ for all $x, y \in K$.

Definition: An ideal of a Lie algebra L is a subspace $I \subseteq L$ such that $[x, y] \in I$, for all $x \in L$, for all $y \in I$.

Definition: Let L be a Lie algebra. The *center* of L is the Lie subalgebra

$$Z(L) := \{x \in L : [x, y] = 0, \forall y \in L\}$$

Definition: Given Lie algebras L_1 and L_2 over a field F , a map $\varphi : L_1 \rightarrow L_2$ is called a *homomorphism*

if φ is linear and $\varphi([x, y]) = [\varphi(x), \varphi(y)]$ for all $x, y \in L_1$.

Definition: Given a Lie algebra L , we define the *adjoint homomorphism*

$$\begin{aligned}\text{ad} : L &\rightarrow \text{gl}(L) \\ x &\mapsto \text{ad } x\end{aligned}$$

where $\text{ad } x : L \rightarrow L$ is given by

$$\text{ad } x(y) = [x, y]$$

Definition: An *algebra over a field F* is a vector space A over F equipped with a bilinear product $[-, -] : A \times A \rightarrow A$. An algebra is *unital* if it has an identity element with respect to multiplication. It is *associative* if $(xy)z = x(yz)$

Definition: Let A be an algebra over a field F . A derivation of A is an F -linear map $D : A \rightarrow A$ such that

$$D(ab) = aD(b) + D(a)b$$

for all $a, b \in A$.

2 Nilpotent Maps and the Invariance Lemma

Let V be an n -dimensional vector space over a field F . A linear map $T : V \rightarrow V$ is called *nilpotent* if there exists an integer $k \in \mathbb{Z}^{\geq 0}$ such that $T^k = 0$.

Lemma 1. Let L be a Lie subalgebra of $\mathfrak{gl}(V)$ and $x \in L$. If the linear map $x : V \rightarrow V$ is nilpotent, then $\text{ad } x : L \rightarrow L$ is also nilpotent.

Proof. Assume $x \in L$. Notice that for $y \in L$,

$$(\text{ad } x)^m(y) = [x, [x, \dots [x, y] \dots]]$$

is a sum of the terms of the form $x^j y x^{m-j}$ for some j between 0 and m . Suppose $x^r = 0$, then take $m = 2r$, in which case each term of the sum is equal to zero. Thus $(\text{ad } x)^{2r} = 0$. $\square$

We now extend the notion of eigenvalues and eigenvectors of a linear map to that of a family of linear maps. Let A be a subalgebra of $\mathfrak{gl}(V)$. We say v is an *eigenvector for A* if v is an eigenvector for every element of A , ie, $a(v) \in \text{span}(v)$ for all $a \in A$. For example, if $A = d(n, F)$, the Lie subalgebra consisting of diagonal matrices, then each standard basis element e_i of F^n is an eigenvector for A .

The notion of eigenvalue for a family of linear transformations is not as easy to generalize. Since the eigenvalue for a single eigenvector v may vary from linear transformation to a different one, we think of eigenvalues as functions $\lambda : A \rightarrow F$ such that $a(v) = \lambda(a)v$ for all $a \in A$. We make the following definition.

Definition. A *weight* for a Lie sub-algebra A of $\mathfrak{gl}(V)$ is a linear map $\lambda : A \rightarrow F$ such that

$$V_\lambda := \{v \in V : a(v) = \lambda(a)v \quad \forall a \in A\}$$

is a non-zero subspace of V . The vector space V_λ is called the *weight space* associated to the weight λ .

Example Let $A = b(n, F)$ be the Lie subalgebra of $\mathfrak{gl}(n, F)$ of upper triangular matrices. Then, e_1 is easily seen to be an eigenvector for A . The corresponding weight is the function $\lambda : A \rightarrow F$ given by $\lambda(a) = a_{11}$ with weight space $\text{span}(e_1)$.

2.1 The Invariance Lemma

We want to generalize the notion of being W -invariant to Lie subalgebras. We start with a simple lemma.

Lemma 1. Suppose A is an ideal of a Lie subalgebra L of $\mathfrak{gl}(V)$. Let

$$W = \{v \in V : a(v) = 0 \quad \forall a \in A\}$$

Then, W is an L -invariant subspace of V .

Proof. Let $w \in W, x \in L$. Then,

$$a(xw) = x(aw) + [a, x]w = 0$$

$x(aw) = 0$ by the fact that $w \in W$. On the other hand, A is an ideal and so $[a, x] \in A$ then $[a, x]w = 0$ too. $\square$

We now state the main lemma.

The Invariance Lemma

Assume that F has characteristic zero. Let L be a Lie subalgebra of $\mathfrak{gl}(V)$ and let A be an ideal of L . Let $\lambda : A \rightarrow F$ be a weight of A . The associated weight space

$$V_\lambda = \{v \in V : av = \lambda(a)v \quad \forall a \in A\}$$

Proof. Let $y \in L$ and $w \in V_\lambda$. Our goal is to show $y(w) \in V_\lambda$, that is, $y(w)$ is an eigenvector for every element of A , with eigenvalue of $a \in A$ given by $\lambda(a)$. Let $a \in A$. Notice that we have

$$a(yw) = y(aw) + [a, y](w) = \lambda(a)yw + \lambda([a, y])w$$

This manipulation shows that we need to prove that the eigenvalue of the commutator $[a, y]$ on V_λ is zero.

Let $U = \text{span}(w, y(w), y^2(w), \dots)$. Since V is finite dimensional and $U \subseteq V$, we can let m be the least number such that the vectors $w, y(w), \dots, y^m(w)$ are linearly dependent. Then, U is m -dimensional and has as a basis

$$w, y(w), \dots, y^{m-1}(w)$$

Now, we claim that if $z \in A$, then z has an upper triangular matrix representation with respect to the basis above whose diagonal entries equal to $\lambda(z)$. We prove this by induction on the number of the column. Since $zw = \lambda(z)w$, the first column of the matrix fits upper triangular conditions. Next, since $[z, y] \in A$, we have

$$z(yw) = y(zw) + [z, y]w = \lambda(z)y(w) + \lambda([z, y])w$$

So the second column has $\lambda([z, y])$ as the first entry and $\lambda(z)$ as the second entry and the rest is 0. For column r , we have

$$z(y^r(w)) = zy(y^{r-1}w) = (yz + [z, y])y^{r-1}w$$

□

3 Representation Theory Review

We will seek new ways to view an abstract Lie algebra. In particular, we are interested in approaching Lie algebras as a subalgebra of the endomorphism algebra of a finite dimensional vector space. We start with the following definition.

Let L be a Lie algebra over a field F . A *representation* of L is a Lie algebra homomorphism

$$\varphi : L \rightarrow \mathfrak{gl}(V)$$

where V is a finite-dimensional vector space over F . One usually omits the homomorphism and says V is a representation of L .

This point of view allows one to fix a basis for V and write the linear transformation afforded by elements of L in matrix form. It is also possible to specify a representation by providing a homomorphism $L \rightarrow \mathfrak{gl}(n, F)$, in which case the representation is called a *matrix representation*.

Given a representation $\varphi : L \rightarrow \mathfrak{gl}(V)$ of L , one can consider the information transferred to $\mathfrak{gl}(V)$ via φ . If $\ker \varphi = 0$, then $\varphi(L)$ encapsulates all information on the structure of L . In this case, the representation is said to be *faithful*.

3.1 Some Examples of Representations

- The adjoint map

$$\begin{aligned} \text{ad} : L &\rightarrow \mathfrak{gl}(L) \\ \text{ad } x(y) &= [x, y] \end{aligned}$$

is a Lie algebra homomorphism which provides a representation of L with $V = L$. This representation is called the *adjoint representation*. Since $\ker \text{ad} = Z(L)$, the adjoint representation is faithful whenever $Z(L) = 0$. This is precisely the case when $L = \mathfrak{sl}(2, \mathbb{C})$.

- If L is a Lie subalgebra of $\mathfrak{gl}(V)$, then the inclusion map $\iota : L \rightarrow \mathfrak{gl}(V)$ induces the *natural representation* of L . One can easily observe that the natural representation is always faithful.
- Given a Lie algebra L over F , let $V = F$ and define $\varphi(x) = 0$ for all $x \in L$. This representation is called the *trivial representation* and is never faithful for non-zero L .

3.2 Introducing Modules for Lie Algebras

Sometimes we are more interested in understanding the vector space on which a given Lie algebra acts than studying the structure of the Lie algebra itself. We achieve this shift in perspective through *modules*.

Definition. Suppose L is a Lie algebra over a field F . A *Lie module* for L , or alternatively an L -*module*, is a finite dimensional F -vector space V together with a map

$$L \times V \rightarrow V \quad (x, v) \mapsto x \cdot v$$

satisfying the conditions

$$\begin{aligned} (\lambda x + \mu y) \cdot v &= \lambda(x \cdot v) + \mu(y \cdot v) \\ x \cdot (\lambda v + \mu w) &= \lambda(x \cdot v) + \mu(x \cdot w) \\ [x, y] \cdot v &= x \cdot (y \cdot v) - y \cdot (x \cdot v) \end{aligned}$$

for all $x, y \in L$, $v, w \in V$ and $\lambda, \mu \in F$.

It is worthwhile to note that the elements of L act on V as linear maps. This hints at the fact that Lie modules and representations are essentially two different ways of describing the same structures. Given a representation $\varphi : L \rightarrow \mathfrak{gl}(V)$, one can turn V into an L -module by defining $x \cdot v := \varphi(x)(v)$ for all $x \in L, v \in V$. Conversely, given a L -module V , one can view V as a representation of L with the homomorphism $\varphi : L \rightarrow \mathfrak{gl}(V)$ given by $\varphi(x)(v) := x \cdot v$.

3.3 Submodules and Factor Modules

As any other object in algebra, modules also enjoy submodule and factor module constructions. We make this notion precise in this subsection.

Definition. Given a Lie module V for some Lie algebra L , a *submodule* of V is defined to be a subspace W of V which is invariant under the action of L . That is, for each $w \in W$ and $x \in L$, we have $x \cdot w \in W$.

Example 1. Let L be a Lie algebra. We may make L into an L -module via the adjoint representation. We show that the submodules of V are precisely the ideals of L . Let $I \subset L$ be an ideal of L . Let $x \in L, v \in I$. Then, $x \cdot v = \text{ad } x(v) = [x, v] \in I$. Thus, I is ad -invariant subspace of L and therefore a submodule of L . Let $V \subset L$ be a submodule of L . Then, $\forall x \in L, v \in V, x \cdot v = [x, v] \in V$. By definition, V is an ideal of L .

Example 2. Let $L = b(n, F)$ be the Lie algebra of $n \times n$ upper triangular matrices and let V be the natural L -module, ie $V = F^n$ and the action of L is given by applying the matrices to column vectors. Let $e_1, \dots, e_n$ be the standard basis of F^n . For $1 \leq r \leq n$, we show that $W_r = \text{span}(e_1, \dots, e_r)$ is a submodule of V . Notice that for $x = (x_{ij}) \in L$ and $e_i \in \{e_1, \dots, e_r\}$. Then,

$$\begin{aligned} x \cdot e_i &= \begin{pmatrix} x_{1i} \\ \vdots \\ x_{ii} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \\ &= x_{1i} \cdot e_1 + \dots + x_{ii} \cdot e_i \in W_r \end{aligned}$$

Thus, the basis elements of W_r are again sent to W_r by the action of L . This shows W_r is invariant under the action of L and therefore a submodule of V .

Suppose W is a submodule of the L -module V . We equip the quotient vector space V/W with an L -module structure by setting

$$x \cdot (v + W) = x \cdot v + W \quad \forall x \in L, v \in V$$

The module obtained this way is called *the quotient or factor module* V/M .

Example 3. Let I be an ideal of the Lie algebra L . Then, by Example 1, I is a submodule of L when L is regarded as an L -module via the adjoint representation. Then, the factor module L/I becomes an L -module by setting

$$x \cdot (y + I) := \text{ad } x(y) + I = [x, y] + I$$

This provides a different way to view the quotient *algebra* L/I . Namely, the factor module L/I is the adjoint representation of the factor algebra L/I on itself.

3.4 Irreducible and Indecomposable Modules

Definition. A Lie module V is said to be *irreducible* or *simple* if it is non-zero and it has no submodules other than $\{0\}$ and V .

Examples

- If V is 1-dimensional, then V is irreducible.
- If L is a simple Lie algebra, then L viewed as an L -module via the adjoint representation is irreducible.

There is a useful criterion that determines whether or not a given module V is irreducible. **Lemma** An L -module V is irreducible if and only if for any non-zero $v \in V$ the submodule generated by v contains all elements of V , where this submodule is defined to be the subspace of V spanned by all elements of the form

$$x_1 \cdot (x_2 \cdot \dots \cdot (x_m \cdot v) \dots)$$

where $x_1, \dots, x_m \in L$.

Proof. Assume V is irreducible. Let $\langle x \rangle$ denote the submodule generated by x . Then, we must have $\langle x \rangle = V$ by definition of irreducibility. Conversely, assume $\langle x \rangle = V$ for all $x \in V$. Then, let $W \subset V$ be a non-zero submodule. Pick a nonzero vector $v \in W$. Since $v \in W$, W is invariant under the action of L , we must have $V = \langle v \rangle \subset W \implies W = V$. $\square$

Definition. Let V be an L -module. If $V = U \oplus W$ where both U and W are L -submodules of V , then V is the *direct sum* of U and W . V is said to be *indecomposable* if there are no non-zero submodules U and W such that $V = U \oplus W$. An irreducible module is indecomposable but the converse is not true.

An L -module V is *completely reducible* if it can be written as a direct sum of irreducible L -modules; that is, $V = S_1 \oplus S_2 \oplus \dots \oplus S_k$ where each S_i is an irreducible L -module.

Example 1. Let F be a field and $L = d(n, F)$, the subalgebra of $\mathfrak{gl}(n, F)$ consisting of diagonal matrices. The natural module $V = F^n$ is completely reducible. In particular, we can write $V = S_1 \oplus \dots \oplus S_k$ where $S_i = \text{span}(e_i)$

3.5 Homomorphisms and Schur's Lemma

The goal of this subsection is to understand the homomorphisms between irreducible modules.

Definition. Let L be a Lie-algebra and let V and W be L -modules. An *L -module homomorphism* or *Lie homomorphism* from V to W is a linear map $\psi : V \rightarrow W$ such that

$$\psi(x \cdot v) = x \cdot \psi(v)$$

for all $x \in L, v \in V$. As is the case for every linear map, L -module isomorphisms give rise to isomorphism theorems which we summarize below.

Isomorphism Theorems

- Let $\psi : V \rightarrow W$ be a homomorphism of L -modules. Then $\ker \psi$ is an L -submodule of V and there is an isomorphism of L -modules

$$V/\ker \psi \cong \text{im } \psi$$

- If U and W are submodules of V , then $U + W$ and $U \cap W$ are submodules of V and $(U + W)/W \cong U/(U \cap W)$.
- If U and W are submodules of V such that $U \subseteq W$, then W/U is a submodule of V/U and the factor module $(V/U)/(W/U)$ is isomorphic to V/W .
- If U is a submodule of V , then there is a bijective correspondence between the submodules of V/U and the submodules of V containing U .

Proof. Let $\theta : V/\ker \psi \rightarrow W$ be given by

$$\theta(v + \ker \psi) = \psi(v)$$

This map is well-defined since if $v + \ker \psi = v' + \ker \psi$ then $v - v' \in \ker \psi$ and

$$\begin{aligned} \theta(v + \ker \psi) &= \psi(v) \\ &= \psi(v - v') + \psi(v') \\ &= \psi(v') \\ &= \theta(v' + \ker \psi) \end{aligned}$$

V and W are vector spaces, which are also additive groups. By isomorphism theorems for groups, we already have that $V/\ker \psi \cong \text{im } \psi$ as groups. Now, we check that θ is an L -module isomorphism.

$$\begin{aligned} \theta(x \cdot (v + \ker \psi)) &= \theta(x \cdot v + \ker \psi) \\ &= \psi(x \cdot v) \\ &= x \cdot \psi(v) \\ &= x \cdot \theta(v + \ker \psi) \end{aligned}$$

This proves the first isomorphism theorem.

Let U, W be sub-modules of V . Define the map

$$\begin{aligned}\varphi : U + W/W &\rightarrow U/U \cap W \\ (u + w) + W &\mapsto u + (U \cap W)\end{aligned}$$

First, we show φ is well-defined. If $(u + w) + W = (u' + w') + W$, then $u - u' \in W$. Then,

$$\begin{aligned}\varphi((u + w) + W) &= u + W \\ &= u' + (u - u') + W \\ &= u' + W \\ &= \varphi((u' + w') + W)\end{aligned}$$

As additive groups, we have $(U + W)/W \cong U/U \cap W$ already. Let's check that φ is an L -module homomorphism. We have

$$\begin{aligned}\varphi(x \cdot ((u + w) + W)) &= \varphi(x \cdot u + x \cdot w + W) \\ &= x \cdot u \\ &= r \cdot \varphi((u + w) + W)\end{aligned}$$

This proves the second isomorphism theorem.

Let U, W be submodules of V with $U \subseteq W$. As groups, $(V/U)/(W/U) \cong V/W$. Consider the map

$$\begin{aligned}\varphi : (V/U)/(W/U) &\rightarrow V/W \\ (v + U) + W/U &\mapsto v + W\end{aligned}$$

We first show that φ is well-defined. If $(v + U) + W/U = (v' + U) + W/U$ then $(v + U) - (v' + U) = v - v' + U \in W/U \implies v - v' \in W$. Then,

$$\begin{aligned}\varphi((v + U) + W/U) &= v + W \\ &= v' + W \\ &= \varphi((v' + U) + W/U)\end{aligned}$$

Finally, we check that φ is an L -module homomorphism.

$$\begin{aligned}\varphi(r \cdot ((v + U) + W/U)) &= \varphi((r \cdot v + U) + W/U) \\ &= r \cdot v + W \\ &= r \cdot \varphi((v + U) + W/U)\end{aligned}$$

This proves the third isomorphism theorem. □

Let L be the 1-dimensional Lie algebra spanned by some element $x \in L$. We may define a representation of L on a vector space V by mapping x to any element of $\text{gl}(V)$. Let W be another vector space and let $f : V \rightarrow V$ and $g : W \rightarrow W$ be linear maps. Then, the representation $\varphi_V : L \rightarrow \text{gl}(V)$ given by $\varphi_V(x) = f$ is isomorphic to the representation $\varphi_W : L \rightarrow \text{gl}(W)$ given by $\varphi_W(x) = g$ if and only if there is a vector space isomorphism $\theta : V \rightarrow W$ such that $\theta \circ f = g \circ \theta \implies \theta f \theta^{-1} = g$. Thus, the representations are isomorphic if and only if there are bases of V and W in which f and g are represented by the same matrix.

Now, we try to study the nature of the structure of a module for a Lie algebra by looking at the homomorphisms between it and other modules. We start with an easy lemma.

Lemma. There are no non-zero homomorphisms between non-isomorphic irreducible modules.

Proof. Suppose S and T are irreducible Lie modules and $\theta : S \rightarrow T$ is a non-zero module homomorphism. Then $\text{im } \theta$ is a submodule of T . Since T is irreducible, we must have $\text{im } \theta = T$. Similarly, $\ker \theta$ is a submodule of S , and S is irreducible; hence, we must have $\ker \theta = 0$. This proves θ is an isomorphism. □

We now consider homomorphisms from an irreducible module to itself.

Schur's Lemma

Let L be a complex Lie algebra and let S be a finite-dimensional irreducible L -module. A map $\theta : S \rightarrow S$ is an L -module homomorphism if and only if θ is a scalar multiple of the identity transformation. That is, $\theta = \lambda \cdot I_V$ for some scalar $\lambda \in \mathbb{C}$.

Proof. If $\theta = \lambda \cdot I_V$, then

$$\begin{aligned}\theta(x \cdot v) &= \lambda(x \cdot v) \\ &= x \cdot (\lambda v) \\ &= x \cdot \theta(v)\end{aligned}$$

Thus, θ is an L -module homomorphism. Conversely, suppose that $\theta : S \rightarrow S$ is an L -module homomorphism. Then θ is a linear map between complex vector spaces and thus, it has an eigenvalue, say λ . Since λI_V is an L -module homomorphism, the map $\theta - \lambda I_V$ is also an L -module homomorphism. $\ker(\theta - \lambda I_V)$ is a non-zero submodule of S . Since S is irreducible, we must have $\ker(\theta - \lambda I_V) = S$ and therefore, $\theta = \lambda I_V$ as desired. $\square$

An application of Schur's Lemma. Let L be a complex Lie algebra and let V be an irreducible L -module. If $z \in Z(L)$, then z acts by scalar multiplication on V , ie there is some $\lambda \in \mathbb{C}$ such that $z \cdot v = \lambda v$ for all $v \in V$.

Proof. Notice that the map $v \mapsto z \cdot v$ is an L -module homomorphism since for $x \in L$ and $v \in V$, one has

$$\begin{aligned}z \cdot (x \cdot v) &= x \cdot (z \cdot v) + [z, x] \cdot v \\ &= x \cdot (z \cdot v)\end{aligned}$$

By Schur's lemma, this map is of the form λI_V for some $\lambda \in \mathbb{C}$. Then, $z \cdot v = \lambda v$ for all $v \in V$, proving z acts by scalar multiplication on V . $\square$

Corollary. Irreducible modules for an Abelian Lie algebra over $\mathbb{C}$ are 1-dimensional.

Proof. Suppose that V is an irreducible module for the abelian Lie algebra L . By the previous application of Schur's lemma, we deduce that every element of L acts by scalar multiplication on V . So, the submodule spanned by a single element of V is 1-dimensional. Since V is irreducible, such a submodule must be all of V and therefore V is 1-dimensional as well. $\square$

4 Representations of $\mathfrak{sl}(2, \mathbb{C})$

Our goal is to study the irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$. We fix the following basis for $\mathfrak{sl}(2, \mathbb{C})$

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

4.1 The Modules V_d

Let $\mathbb{C}[x, y]$ denote the vector space of polynomials in two variables with complex coefficients. For each integer $d \geq 0$, let V_d be the subspace of homogeneous polynomials in x and y of degree d . Then, $x^d, x^{d-1}y, x^{d-2}y^2, \dots, xy^{d-1}, y^d$ is a basis for the vector space V_d . Consider the following map

$$\begin{aligned} \varphi : \mathfrak{sl}(2, \mathbb{C}) &\rightarrow \mathfrak{gl}(V_d) \\ \varphi(e) &:= x \frac{\partial}{\partial y} \\ \varphi(f) &:= y \frac{\partial}{\partial x} \\ \varphi(h) &:= x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} \end{aligned}$$

Once proven to be a representation of $\mathfrak{sl}(2, \mathbb{C})$, φ makes V_d into a $\mathfrak{sl}(2, \mathbb{C})$ -module. Notice that $\varphi(h)(x^a y^b) = (a - b)x^a y^b$. Therefore, h acts diagonally on V_d with respect to our chosen basis.

Theorem The map φ defined above is a representation of $\mathfrak{sl}(2, \mathbb{C})$.

Proof. By construction, φ is linear. We check that φ preserves the Lie bracket on the basis elements.

Let's first confirm that $[\varphi(e), \varphi(f)] = \varphi([e, f]) = \varphi(h)$. We apply this map to a basis element $x^a y^b$ of V_d .

$$\begin{aligned} [\varphi(e), \varphi(f)](x^a y^b) &= \varphi(e)(\varphi(f)(x^a y^b)) - \varphi(f)(\varphi(e)(x^a y^b)) \\ &= \varphi(e)(ax^{a-1}y^{b+1}) - \varphi(f)(bx^{a+1}y^{b-1}) \\ &= a(b+1)x^a y^b - b(a+1)x^a y^b \\ &= (a-b)x^a y^b \\ &= \varphi(h)(x^a y^b) \\ &= \varphi([e, f])(x^a y^b) \end{aligned}$$

Similarly, we check that $[\varphi(f), \varphi(h)] = \varphi([f, h]) = 2\varphi(f)$

$$\begin{aligned} [\varphi(f), \varphi(h)](x^a y^b) &= \varphi(f)((a-b)x^a y^b) - \varphi(h)(ax^{a-1}y^{b+1}) \\ &= a(a-b)x^{a-1}y^{b+1} - a(a-1)x^{a-1}y^{b+1} + (b+1)ax^{a-1}y^{b+1} \\ &= 2ax^{a-1}y^{b+1} \\ &= 2\varphi(f)(x^a y^b) \end{aligned}$$

Finally, we check $[\varphi(h), \varphi(e)] = \varphi([h, e]) = \varphi(2e) = 2\varphi(e)$.

$$\begin{aligned} [\varphi(h), \varphi(e)](x^a y^b) &= \varphi(h)(bx^{a+1}y^{b-1}) - \varphi(e)((a-b)x^a y^b) \\ &= b(a+1)x^{a+1}y^{b-1} - b(b-1)x^{a+1}y^{b-1} - b(a-b)x^{a+1}y^{b-1} \\ &= 2bx^{a+1}y^{b-1} \\ &= 2\varphi(e)(x^a y^b) \end{aligned}$$

□

Since φ is a representation, each basis element corresponds to a matrix. We have the following matrix

representations for $\varphi(e)$, $\varphi(f)$ and $\varphi(h)$ with respect to the basis $x^d, x^{d-1}y, \dots, y^d$ of V_d .

$$\begin{aligned}\varphi(e) &= \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} \\ \varphi(f) &= \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ d & 0 & \dots & 0 & 0 \\ 0 & d-1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix} \\ \varphi(h) &= \begin{pmatrix} d & 0 & \dots & 0 & 0 \\ d & d-2 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -d+2 & 0 \\ 0 & 0 & \dots & 0 & -d \end{pmatrix}\end{aligned}$$

Theorem 1 The $\mathfrak{sl}(2, \mathbb{C})$ -submodule of V_d generated by any particular basis element $x^a y^b$ contains all the basis elements and so is all of V_d .

Proof. Let $x^a y^b$ be a basis element. Then, applying $\varphi(f)$ to $x^a y^b$ k times will result in $\frac{(d-a)!}{(d-a-k)!} x^{a-k} y^{b+k}$. At that point, we can apply a suitable multiple of $\varphi(h)$ to get $x^{a-k} y^{b+k}$. Similar argument with $\varphi(e)$ and $\varphi(h)$ shows we can also get $x^{a+k} y^{b-k}$, proving the submodule generated by $x^a y^b$ is all of V_d . $\square$

Theorem 2. The $\mathfrak{sl}(2, \mathbb{C})$ -module V_d is irreducible.

Proof. Assume U is a non-zero $\mathfrak{sl}(2, \mathbb{C})$ -submodule of V_d . Then $h \cdot u \in U$ for all $u \in U$. Since h acts diagonalisably on V_d , it also acts diagonalisably on U , so there is an eigenvector of h which lies in U . The eigenspaces of h on V_d are 1-dimensional and hence spanned by some basis element $x^a y^b$. By Theorem 1, the submodule generated by $x^a y^b$ is all of V_d , which implies $V_d \subset U$. Thus, $U = V_d$ as desired. $\square$

4.2 Classifying the Irreducible $\mathfrak{sl}(2, \mathbb{C})$ -modules

For different d , the $\mathfrak{sl}(2, \mathbb{C})$ -modules V_d are clearly not isomorphic. The aim of this subsection is to prove that any finite-dimensional irreducible $\mathfrak{sl}(2, \mathbb{C})$ module is isomorphic to one of the V_d . We start with a simple lemma.

Lemma 1 . Suppose that V is an $\mathfrak{sl}(2, \mathbb{C})$ -module and $v \in V$ is an eigenvector of h with eigenvalue λ . Then,

- either $e \cdot v = 0$ or $e \cdot v$ is an eigenvector of h with eigenvalue $\lambda + 2$.
- Either $f \cdot v = 0$ or $f \cdot v$ is an eigenvector of h with eigenvalue $\lambda - 2$.

Proof. We have

$$\begin{aligned}h \cdot (e \cdot v) &= e \cdot (h \cdot v) + [h, e] \cdot v \\ &= e \cdot (\lambda v) + 2e \cdot v \\ &= (\lambda + 2)e \cdot v\end{aligned}$$

This proves the first assertion in the lemma. Moreover,

$$\begin{aligned}h \cdot (f \cdot v) &= f \cdot (h \cdot v) + [h, f] \cdot v \\ &= f \cdot (\lambda v) - 2f \cdot v \\ &= (\lambda - 2)f \cdot v\end{aligned}$$

proving the second assertion. $\square$

We need another lemma before tackling the main result.

Lemma 2. Let V be a finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$ -module. Then V contains an eigenvector w for h such that $e \cdot w = 0$.

Proof. The linear map $h : V \rightarrow V$ has at least one eigenvalue and thus at least one eigenvector. Let $h \cdot v = \lambda v$. Consider the set of vectors

$$v, e \cdot v, e^2 \cdot v, \dots$$

If all are non-zero, then by Lemma 1, they form an infinite sequence of h -eigenvectors with distinct eigenvalues. Hence, each vector in this set would be independent, contradicting the fact that V is finite dimensional. Thus, there exists $k \in \mathbb{N}$ such that $e^k \cdot v \neq 0$ and $e^{k+1} \cdot v = 0$. Set $w = e^k \cdot v$. Then, $h \cdot w = (\lambda + 2k)w$ and $e \cdot w = 0$. $\square$

The Main Theorem

If V is a finite-dimensional irreducible $\mathfrak{sl}(2, \mathbb{C})$ -module, then V is isomorphic to one of the V_d .

Proof. By the previous lemma, we know that v has an h -eigenvector w such that $e \cdot w = 0$. Now consider the sequence

$$w, f \cdot w, f^2 \cdot w, \dots$$

Again, by Lemma 2, there exists $d \geq 0$ with $f^d \cdot w \neq 0$ and $f^{d+1} \cdot w = 0$. We prove the theorem in three steps.

Step 1. The vectors $w, f \cdot w, \dots, f^d \cdot w$ form a basis for a submodule of V .

Proof. The Lemma 1, they are eigenvectors for h with distinct eigenvalues and hence linearly independent. By construction, their span is invariant under h and f . We use induction on k to prove their span is also invariant under e . More precisely, we need to prove that

$$e \cdot (f^k \cdot w) \in \text{span}(f^j \cdot w : 0 \leq j < k)$$

If $k = 0$, then $e \cdot w = 0$ so there is nothing to prove. Note that we have

$$[\varphi(e), \varphi(f)] = \varphi(h) \iff ef - fe = h$$

Therefore

$$\begin{aligned} e \cdot (f^k \cdot w) &= (ef) \cdot (f^{k-1} \cdot w) \\ &= (fe + h) \cdot (f^{k-1} \cdot w) \end{aligned}$$

By the inductive hypothesis, $f^{k-1} \cdot w \in \text{span}(f^j \cdot w : 0 \leq j < k-1)$. Therefore $fe f^{k-1} \cdot w \in \text{span}(f^j \cdot w : 0 \leq j < k)$. Moreover, $h f^{k-1} \cdot w = c \cdot f^{k-1} \cdot w$ for some scalar $c \in \mathbb{C}$. Thus, $e \cdot (f^k \cdot w) \in \text{span}(f^j \cdot w : 0 \leq j < k)$. Since V is irreducible, the submodule spanned by $f^k \cdot w$ for $0 \leq k \leq d$ is equal to V . $\square$

Step 2. We claim that $\lambda = d$.

Proof. By Lemma 1, we know that the matrix of h with respect to the basis $w, f \cdot w, \dots, f^d \cdot w$ is diagonal and the trace of that matrix is

$$\lambda + (\lambda - 2) + \dots + (\lambda - 2d) = (d+1)\lambda - (d+1)d$$

Since $[e, f] = h$ and $\text{tr}[e, f] = \text{tr}(ef - fe) = 0$, we have $(d+1)\lambda - (d+1)d = 0 \implies \lambda = d$. $\square$

Step 3. Final step is to provide an explicit isomorphism $\psi : V \rightarrow V_d$. V has basis $\{w, f \cdot w, \dots, f^d \cdot w\}$. Furthermore, V_d has basis

$$\{x^d, f \cdot x^d, \dots, f^d \cdot x^d\}$$

where $f^k \cdot x^d$ is a scalar multiple of $x^{d-k}y^k$. This result can easily be seen from the diagram representation of the action of h, e, f on V_d , which we provide below. Also notice that the eigenvalue of h on $f^k \cdot w$ and the eigenvalue of h on $f^k \cdot x^d$ are both equal to $d - 2k$. We may simply set

$$\psi(w) = x^d$$

and then define $\psi : V \rightarrow V_d$ by

$$\psi(f^k \cdot w) := f^k \cdot x^d$$

This vector space isomorphism commutes with the actions of f and h by construction. We use induction on k to prove ψ also commutes with the action of e . For $k = 0$, we get $\psi(e \cdot w) = 0$ and $e \cdot \psi(w) = e \cdot x^d = 0$. For the inductive step, we have that

$$\psi(e f^k \cdot w) = \psi((fe + h) \cdot (f^{k-1} \cdot w)) = f \cdot \psi(e f^{k-1} \cdot w) + h \cdot \psi(f^{k-1} \cdot w)$$

By the induction hypothesis, $\psi(e f^{k-1} \cdot w) = e \cdot \psi(f^{k-1} \cdot w)$. Using $h = [e, f]$, we get

$$\begin{aligned} \psi(e f^k \cdot w) &= (fe + h) \cdot \psi(f^{k-1} \cdot w) \\ &= fe \cdot \psi(f^{k-1} \cdot w) + h \cdot \psi(f^{k-1} \cdot w) \\ &= (fe + [e, f]) \cdot \psi(f^{k-1} \cdot w) \\ &= e \cdot \psi(f^k \cdot w) \end{aligned}$$

This proves the theorem. □

Corollary If V is a finite-dimensional representations of $\mathfrak{sl}(2, \mathbb{C})$ and $w \in V$ is an h -eigenvector such that $e \cdot w = 0$, then $h \cdot w = dw$ for some non-negative integer d and the submodule of V generated by w is isomorphic to V_d .

Proof. By step 1 of the previous proof, there exists some $d \geq 0$ such that the vectors $w, f \cdot w, \dots, f^d \cdot w$ span a submodule of V . If we apply steps 2 and three to this submodule we end up with an Lie module isomorphism $\psi : \langle w \rangle \rightarrow V_d$ as desired. □

5 Vector Fields on $\mathbb{R}^2$ Generated by Exponentiation

We move on to investigating the structure of vector fields on $\mathbb{R}^2$ generated by exponentiation on this section of the paper. Given a matrix $M \in \mathfrak{sl}(2, \mathbb{C})$, we consider the vector field on $\mathbb{R}^2$ defined by

$$X_p = \left. \frac{d}{dt} \right|_{t=0} e^{Mt} p$$

We start by explicitly constructing the vector field generated by the basis elements of $\mathfrak{sl}(2, \mathbb{C})$.

$M = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ In this case, we have

$$e^{Mt} = I + Mt = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

Therefore, for $p = (x, y) \in \mathbb{R}^2$, we get

$$X_p = \left. \frac{d}{dt} \right|_{t=0} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

This gives the vector field

$$X = y \frac{\partial}{\partial x}$$

on $\mathbb{R}^2$. whose plot is given below.

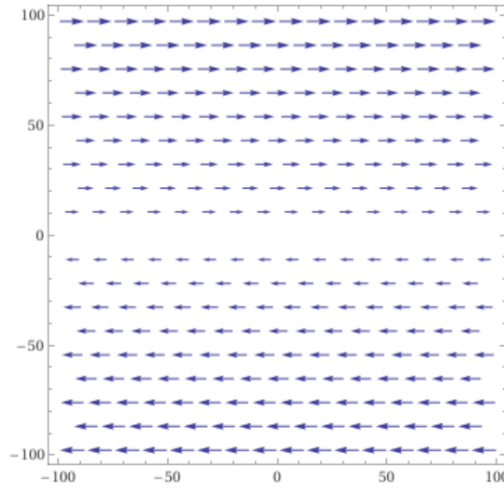


Figure 1: $X = y \frac{\partial}{\partial x}$

$M = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ In this case, we have

$$e^{Mt} = I + Mt = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$$

Therefore, for $p = (x, y) \in \mathbb{R}^2$, we get

$$X_p = \left. \frac{d}{dt} \right|_{t=0} \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ x \end{pmatrix}$$

This gives the vector field

$$X = x \frac{\partial}{\partial y}$$

on $\mathbb{R}^2$. The plot of the vector field is given below.

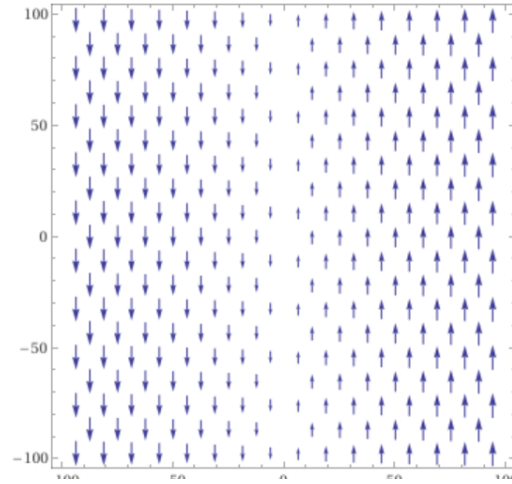


Figure 2: $X = x \frac{\partial}{\partial y}$

$M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ In this case, we have

$$e^{Mt} = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}$$

Therefore, for $p = (x, y) \in \mathbb{R}^2$, we get

$$X_p = \frac{d}{dt} \Big|_{t=0} \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$$

This gives the vector field

$$X = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}$$

on $\mathbb{R}^2$ whose plot is given below.

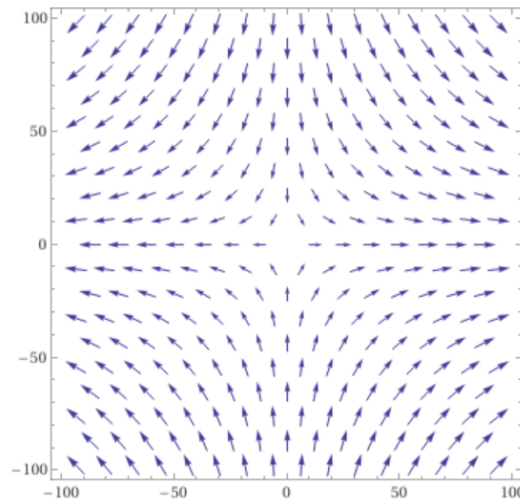


Figure 3: $X = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}$

It is important to notice that these vector fields are invariant under multiplication by e^{Ms} for any $s \in \mathbb{R}$.

As one can see,

$$\begin{aligned}
X_{e^{Ms}p} &= \left. \frac{d}{dt} \right|_{t=0} e^{Mt} e^{Ms} p \\
&= e^{Ms} \left. \frac{d}{dt} \right|_{t=0} e^{Mt} p \\
&= e^{Ms} X_p
\end{aligned}$$

Let E, F, H be the vector fields generated by the given values of M above. That is,

$$\begin{aligned}
X_E &= y \frac{\partial}{\partial x} \\
X_F &= x \frac{\partial}{\partial y} \\
X_H &= x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}
\end{aligned}$$

Let us now compute the Lie bracket of these vector fields.

$$\begin{aligned}
[X_E, X_F] &= \left[y \frac{\partial}{\partial x}, x \frac{\partial}{\partial y} \right] = y \frac{\partial}{\partial y} - x \frac{\partial}{\partial x} = -X_H \\
[X_E, X_H] &= \left[y \frac{\partial}{\partial x}, x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} \right] = 2y \frac{\partial}{\partial x} = 2X_E \\
[X_F, X_H] &= \left[x \frac{\partial}{\partial y}, x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} \right] = -2x \frac{\partial}{\partial y} = -2X_F
\end{aligned}$$

6 Vector Fields on $\mathbb{R}^2$ Generated by Möbius Transformations

We now investigate vector fields on $\mathbb{R}^2$ generated by Möbius transformations. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{C})$. The Möbius transformation given by M is

$$f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$$

$$z \mapsto \frac{az + b}{cz + d}$$

We define a vector field on $\mathbb{R}^2$ by letting e^{Mt} act on p via Möbius transformation. More precisely, given $M \in \mathfrak{sl}(2, \mathbb{C})$. We consider the vector field

$$X_z = \left. \frac{d}{dt} \right|_{t=0} \frac{a(t)z + b(t)}{c(t)z + d(t)}$$

where

$$\begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix} = e^{Mt} \in \mathfrak{sl}(2, \mathbb{C})$$

Notice that since $\det e^A = e^{\text{Tr} A}$ for any matrix A and $\text{Tr} A = 0$ for $A \in \mathfrak{sl}(2, \mathbb{C})$, we see that the exponentiation map $\exp : \mathfrak{sl}(2, \mathbb{C}) \rightarrow SL(2, \mathbb{C})$ is a map from the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ to the Lie group $SL(2, \mathbb{C})$, though it is not surjective.

We first explicitly compute the vector field X given by the basis elements of $\mathfrak{sl}(2, \mathbb{C})$.

$M = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then,

$$e^{Mt} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

Hence the vector field is given by

$$X_z = \left. \frac{d}{dt} \right|_{t=0} (z + t) = 1$$

Interpreted in terms of $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ this vector field is simply

$$X = \frac{\partial}{\partial x}$$

This vector field does not look much interesting.

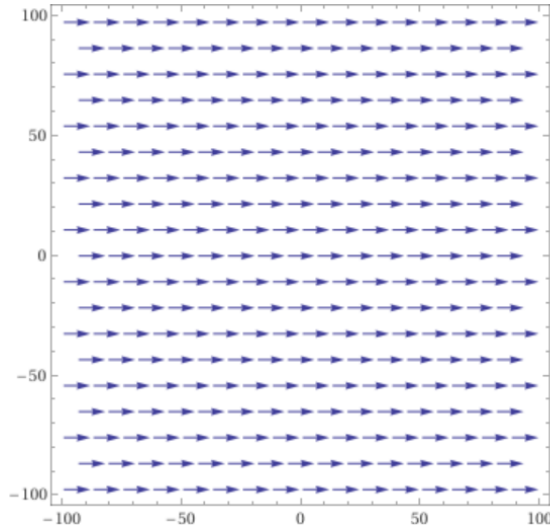


Figure 4: $X = \frac{\partial}{\partial x}$

$\mathbf{M} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then,

$$e^{Mt} = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$$

Hence the vector field is given by

$$X_z = \frac{d}{dt} \Big|_{t=0} \frac{z}{tz + 1} = \frac{d}{dt} \Big|_{t=0} z(1 - tz) = -z^2$$

Writing $z = x + iy$, we can get the real and imaginary parts of the expression, which will correspond to the coefficients of $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$.

$$\begin{aligned} X_z &= -z^2 \\ &= (-x^2 + y^2) \frac{\partial}{\partial x} - 2xy \frac{\partial}{\partial y} \end{aligned}$$

An plot of this vector field is given below.

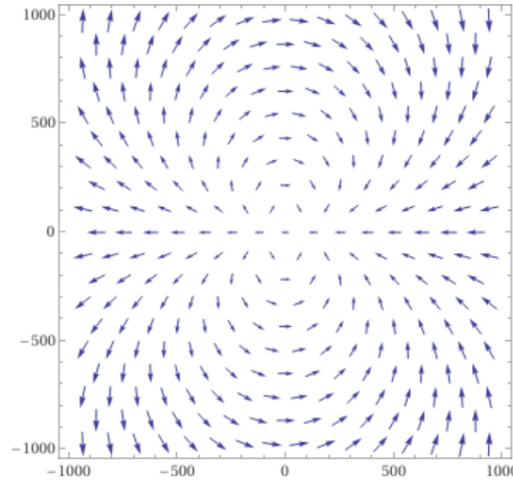


Figure 5: X_F

$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Applying the same procedure, we get

$$e^{Mt} = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}$$

Then,

$$\begin{aligned} X_z &= \frac{d}{dt} \Big|_{t=0} \frac{e^t z}{e^{-t} z} \\ &= \frac{d}{dt} \Big|_{t=0} e^{2t} z \\ &= 2z \end{aligned}$$

Writing $z = x + iy$, the vector field is

$$X_z = 2x \frac{\partial}{\partial x} + 2y \frac{\partial}{\partial y}$$

A plot of the vector field is given below. Let X_E, X_F, X_H be the vector fields computed above, respec-

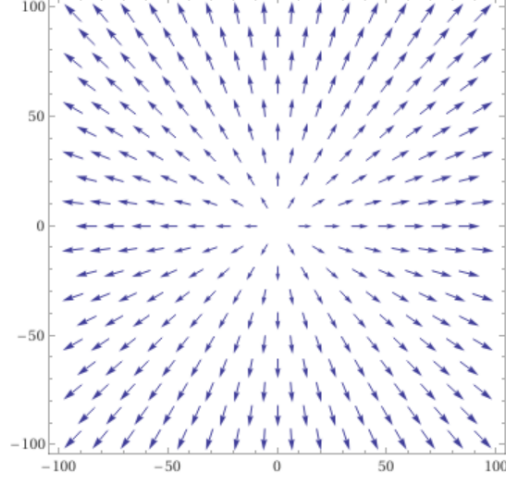


Figure 6: $X = 2x \frac{\partial}{\partial x} + 2y \frac{\partial}{\partial y}$

tively. We are now interested in computing the Lie bracket of these vector fields.

$$\begin{aligned} [X_E, X_H] &= 2 \frac{\partial}{\partial x} = 2X_E \\ [X_E, X_F] &= -2x \frac{\partial}{\partial x} - 2y \frac{\partial}{\partial y} = -X_H \\ [X_F, X_H] &= 2(x^2 - y^2) \frac{\partial}{\partial x} + 4xy \frac{\partial}{\partial y} = -2X_F \end{aligned}$$

In general, if we take $M = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in \mathfrak{sl}(2, \mathbb{C})$ then

$$e^{Mt} = I + tM + O(t^2) = \begin{pmatrix} 1 + ta & tb \\ tc & 1 - ta \end{pmatrix} + O(t^2)$$

and the corresponding vector field is

$$\begin{aligned} X_z &= \left. \frac{d}{dt} \right|_{t=0} \frac{(1 + ta)z + tb}{tcz + 1 - ta} \\ &= \left. \frac{d}{dt} \right|_{t=0} \frac{z + t(az + b)}{1 + t(cz - a)} \\ &= \left. \frac{d}{dt} \right|_{t=0} (z + t(az + b))(1 - t(cz - a)) \\ &= az + b - z(cz - a) \end{aligned}$$

Now, if we write $z = x + iy, a = a_1 + ia_2, b = b_1 + ib_2, c = c_1 + ic_2$, then we get

$$X = (-c_1x^2 + 2a_1x + 2c_2xy - 2a_2 + c_1y^2 + b_1) \frac{\partial}{\partial x} + (-c_2x^2 + 2a_2x - 2c_1xy + 2a_1y + c_2y^2 + b_2) \frac{\partial}{\partial y}$$

One can deduce the previous vector fields by plugging in the right values for a_1, a_2, b_1, b_2, c_1 and c_2 .

References

- [1] Karin Erdmann and Mark J. Wildon. *Introduction to Lie Algebras*. Springer Undergraduate Mathematics Series. London: Springer, 2006.